

Modern-Era Challenges in the Clinical Management of Graft-Versus-Host Disease

Catherine J. Lee and Paul A. Carpenter

Abstract

For several decades, graft-versus-host disease (GVHD) has been a long-standing barrier to successful allogeneic hematopoietic cell transplantation and a significant cause of post-transplant morbidity and mortality. Initially described as secondary disease or wasting syndrome in transplanted mice, the pathobiology of GVHD is increasingly understood as a dynamic interplay between innate and adaptive immunity in response to initial tissue damage, leading to inflammation and end-organ damage. In parallel, more uniform symptom capture, diagnosis, and response criteria have facilitated rigorous clinical trial design and conduct; together, these advancements have facilitated the development of novel GVHD prevention and treatment strategies. While these advancements have improved the GVHD treatment paradigm, new questions arise within this complex patient population. This chapter discusses several of the most pertinent current clinical practice challenges in GVHD, including its earlier diagnosis, risk stratification, initial and more advanced stage management, as well as a

renewed focus on supportive care, given our increased understanding of key roles played by the human microbiome.

Keywords

Hematopoietic cell transplantation · Graft versus host disease · Acute GVHD · Chronic GVHD · Steroid-refractory · Corticosteroids · Immunosuppression · Treatment · Gastrointestinal · Fibrotic · Supportive care · Microbiome · Nutrition

6.1 Introduction

Graft-versus-host disease (GVHD) is a significant complication following allogeneic hematopoietic cell transplantation (HCT) that can be broadly characterized by donor immune cells attacking recipient tissues due to histocompatibility differences (Holtan et al. 2014). Its acute and chronic forms have distinct clinical presentations and management strategies. For decades, GVHD has been the main complication of allogeneic HCT and the leading cause of post-transplant morbidity and nonrelapse mortality (NRM). Significant progress has been made in understanding its pathophysiology and improving diagnostic criteria and clinical staging. This has led to its improved recognition, more uniform symptom capture, and development of

C. J. Lee (✉) · P. A. Carpenter
Fred Hutchinson Cancer Center, Seattle, WA, USA
University of Washington, Seattle, WA, USA
e-mail: cjlee@fredhutch.org

novel prevention and treatment strategies. While GVHD rates are declining, the clinical complexity of acute GVHD (aGVHD) and chronic (cGVHD) syndromes remains a challenge.

This chapter will specifically discuss the key contemporary challenges of early GVHD diagnosis, risk stratification, and treatment strategies for new onset and steroid-refractory GVHD. We will also address challenges in supportive care following a diagnosis of GVHD. While recognizing that the best GVHD strategy is prevention, we defer this discussion and refer readers to Chap. 6.

6.2 Acute GVHD

Acute GVHD is a frequent complication of HCT involving the activation of donor T lymphocytes against host tissues. Despite prophylactic immunosuppression (IST), up to 50% of HCT recipients experience aGVHD and ~14% experience severe aGVHD (grade III–IV) (Martin et al. 2012; Zeiser and Blazar 2017a; Malard et al. 2020). The aGVHD syndrome most frequently targets the skin (maculopapular rash), gastrointestinal (GI) tract (nausea, vomiting, and/or diarrhea), or liver (traditionally with cholestatic liver injury $\pm$ transaminase elevation), alone or in combination (Glucksberg et al. 1974). *Classic* aGVHD develops before 100 days post-HCT but since 2005, *late* aGVHD has been used to categorize aGVHD that occurs after day 100, more often after nonmyeloablative conditioning (Filipovich et al. 2005). Several commonly used aGVHD grading systems include Glucksberg (Glucksberg et al. 1974), International Bone Marrow Transplant Registry (IBMTR) (Rowlings et al. 1997), Minnesota (MacMillan et al. 2002a, 2015), and Mount Sinai Acute GVHD International Consortium (MAGIC) (Harris et al. 2016). MAGIC grading is the most current and detailed approach that is recommended by the EBMT–NIH–CIBMTR Task Force (Table 6.1) (Schoemans et al. 2018a).

The choice of initial therapy depends on aGVHD severity and initial organ involvement (Malard et al. 2023a). Mild (MAGIC grade 1) aGVHD is most often managed by continuing

aGVHD prophylaxis (or resuming or dose-escalating prophylaxis when it was tapering or already discontinued) and adding organ-directed topical corticosteroids (CS). Treatment of grade 2 or higher aGVHD begins with high-dose CS, traditionally at a prednisone-equivalent dose (PED) of 1–2 mg/kg/day (Martin et al. 2012; Penack et al. 2024). The Seattle group has shown that for adults without liver involvement, with <50% body surface area rash and <1 L per day of diarrhea, therapy can safely and effectively begin with a PED of 0.5–1 mg/kg/day, often supported with organ-directed topical CS (Mielcarek et al. 2009, 2015).

While approximately 60% of grade II aGVHD responds to CS, 30–40% of grade III/IV aGVHD is CS-nonresponsive, leading to shortened overall survival (OS) (Deeg 2007; Westin et al. 2011; MacMillan et al. 2010; Yu et al. 2020; Holtan et al. 2022). The incomplete efficacy of prolonged CS therapy is compounded by significant associated opportunistic infections, metabolic and mood disturbances, bone loss, and osteonecrosis.

6.3 Steroid-Refractory Acute GVHD

Steroid-refractory aGVHD (SR-aGVHD) is most often defined as any organ that is progressing 3–5 days after starting 2 mg/kg/day of PED or no response after 7 days (Pavletic and Fowler 2012; Schoemans et al. 2018b). Other poor prognosis aGVHD scenarios include an initial response followed by recurrence, as well as persistent or worsening visceral aGVHD despite skin improvement (MacMillan et al. 2015; Axt et al. 2019). Patients may also experience steroid-dependent aGVHD (SD-aGVHD), defined as the inability to taper PED to <2 mg/kg/day after at least 7 days of successful initial treatment, or recurrent aGVHD in the same or new organs during CS tapering (Schoemans et al. 2018b). Prognosis in SD-aGVHD has been less studied but a retrospective study suggests a similar 2-year OS and NRM compared to steroid-sensitive aGVHD (El Jurdi et al. 2021). However, the prolonged IST

Table 6.1 MAGIC aGVHD target organ staging and overall severity grading

Stage	Skin (active erythema only)	Liver (total bilirubin)	Upper GI	Lower GI (stool output/day)
0	No rash	<2 mg/dL	No or intermittent ^a anorexia or nausea or vomiting	Adult: diarrhea ^b <500 mL/day or <3 episodes/day Child: diarrhea ^b <10 mL/kg/day or <4 episodes/day
1	Rash <25% of BSA	2–3 mg/dL	Persistent ^a anorexia or nausea or vomiting	Adult: diarrhea ^b 500–999 mL/day or 3–4 episodes/day Child: diarrhea ^b 10–19.9 mL/kg/day or 4–6 episodes/day
2	Rash 25–50% of BSA	3.1–6 mg/dL		Adult: diarrhea ^b 1000–1500 mL/day or 5–7 episodes/day Child: diarrhea ^b 20–30 mL/kg/day or 7–10 episodes/day
3	Rash >50% of BSA	6.1–15 mg/dL		Adult: diarrhea ^b >1500 mL/day or >7 episodes/day Child: >diarrhea ^b 30 mL/kg/day or >10 episodes/day
4	Generalized erythroderma (>50% BSA) plus bullous formation and desquamation >5% of BSA	>15 mg/dL		Severe abdominal pain with or without ileus or grossly bloody stool (regardless of stool volume)

Overall clinical severity grade (based on most severe target organ involvement)

0 No organ involvement (skin = 0; and liver = 0; and GI=0) corresponds to the absence of aGVHD

I Skin = 1 or 2, without liver, upper or lower GI involvement

II Skin = 3; and/or liver = 1; and/or upper GI=1; and/or lower GI=1

III Liver = 2 or 3; and/or lower GI=2 or 3

IV Skin = 4; and/or liver = 4; and/or lower GI=4

Adapted from Schoemans et al. (2018b)

^aTo be suggestive of GVHD, anorexia should be accompanied by weight loss, nausea should last at least 3 days, or be accompanied by at least 2 vomiting episodes per day for at least 2 days

^b One episode of diarrhea is considered to be about 200 mL for an adult and 3 mL/kg for a child (<50 kg)

commonly used in SD-aGVHD is associated with more frequent serious infections (El Jurdi et al. 2021). Last, the inability to tolerate acute high-dose CS toxicities (e.g., uncontrolled infections, hyperglycemia, anxiety) is another indication for second-line therapy (Schoemans et al. 2018b).

Until the US Food and Drug Administration (FDA) approval of the dual JAK1/2 inhibitor, ruxolitinib in May 2019, there was no accepted stan-

dard-of-care treatment for SR-aGVHD (Zeiser and Blazar 2017a). The approval for ruxolitinib rested upon sequential phase 2 (REACH1) (Jagasia et al. 2020) and phase 3 (REACH2) studies (Zeiser et al. 2020) with REACH2 showing superior ~2.6-fold higher overall response rate (ORR) at day 28 for ruxolitinib versus 39% for the “investigator’s choice” control arm of nine commonly available options (ORR 62% vs. 39%); at day 28, the complete response (CR) rate was

34% for ruxolitinib versus 19% for controls. The estimated cumulative 6-month incidence of loss of response was 10% (95% CI, 4–17) for ruxolitinib versus 39% (95% CI, 26–52) for controls. Recipients of ruxolitinib had a significantly higher incidence of thrombocytopenia and a modestly higher incidence of anemia, infection, and cytomegalovirus infection/reactivation compared to recipients of investigator's choice therapy.

Multiple other agents for second-line therapy have been investigated without an identified effective therapy (Martin et al. 2012; Martin 2020). In REACH2, the most common second-line approach was extracorporeal photopheresis (ECP), which accounted for 27% of the investigator's choice arm. Response endpoints for ECP have varied from day 28 to the best overall response, which includes beyond day 28, ranging from 25% to 100% but usually most effective in the skin compared to viscera (Kitko et al. 2022; Drexler et al. 2020). ECP may be preferred over ruxolitinib as second-line therapy for patients with severe infection (Martin 2020).

Despite the landmark FDA approval of ruxolitinib, responses are mostly incomplete and sometimes transient. SR-aGVHD has been associated with high GVHD-associated mortality (Martin et al. 2012; Westin et al. 2011; Abedin et al. 2019; Levine et al. 2010) resulting from cumulative damage to target tissues which can outpace augmented systemic IST attempts, together with a higher incidence of bloodstream infections and other infection-related mortality (Garcia-Cadenas et al. 2017).

Beyond ruxolitinib, there is no established next-line treatment for patients who are intolerant or refractory to ruxolitinib. Enrollment in a clinical trial is recommended but if unavailable, third-line treatment is selected based on center experience with available therapy options and relevance of particular medication safety profiles to individual patients. Treatment options include emerging pharmacologic therapies such as mesenchymal stromal cells (Kebriaei et al. 2020) and fecal microbiota transplantation (Malard et al. 2023b) (Table 6.2), but optimizing treatment for severe SR-aGVHD that has failed second-line therapy remains an unmet medical need.

6.4 Identifying Patients at High Risk of Steroid-Refractory Acute GVHD

Two of the challenges in the clinical management of aGVHD are as follows: (A) identifying who is at high risk (HR) for developing SR-aGVHD and (B) whether more aggressive first-line therapy might mitigate the inferior outcomes observed for HR patients. Conversely, who is more likely to have mild aGVHD, and can this group potentially be spared prolonged high-dose CS? Both the Glucksberg (Glucksberg et al. 1974) and IBMTR (Przepiorka et al. 1995) aGVHD grading approaches fall well short of this challenge. The Minnesota group subsequently developed a refined, clinical-based risk score model, applied at aGVHD onset, that dichotomizes adults (MacMillan et al. 2015) and children (MacMillan et al. 2020) into HR or standard-risk (SR) aGVHD groups who have, respectively, higher and lower risks for NRM. Building on initial observations from 1990 that single-organ involvement, as well as organ severity, were more associated with CR than multiorgan involvement (Weisdorf et al. 1990), the refined Minnesota grading defines SR aGVHD as single-organ involvement (stage 1–3 skin or stage 1–2 GI) or two-organ involvement (stage 1–3 skin plus stage 1 GI; or stage 1–3 skin plus stage 1–4 liver) and is associated with an ORR rate of 69% at day 28 compared to 43% for the HR group, which includes all other patients (MacMillan et al. 2015).

Most recently, the MAGIC algorithm used serum biomarker concentrations of suppressor of tumorigenesis 2 (ST2) and regenerating islet-derived 3 α (REG3 α) to categorize patients into three distinct 6-month NRM risk groups: *low* (Ann Arbor 1), *intermediate* (Ann Arbor 2), and *high* (Ann Arbor 3) (Hartwell et al. 2017; Levine et al. 2015). ST2 and REG3 α both signify gastrointestinal crypt damage (Zhao et al. 2018; Reichenbach et al. 2015) and may identify patients already with, or about to develop, high-risk, lower GI GVHD. In fact, patients without GI symptoms yet deemed high-risk based on Ann Arbor 3 biomarker panel results

Table 6.2 Treatment options for SR-acute GVHD after ruxolitinib

Agent	Function
Alemtuzumab (Gomez-Almaguer et al. 2008; Schnitzler et al. 2009)	Anti-CD52 monoclonal antibody that depletes T and B cells
Alpha-1 antitrypsin (Magenau et al. 2018)	Circulating protease inhibitor with immunomodulatory activity
ATG (MacMillan et al. 2002b)	T-cell depleting antibody
Basiliximab (Schmidt-Hieber et al. 2005)	IL-2 receptor antibody that inhibits T-cell proliferation and activation
CNIs (tacrolimus, cyclosporine)	Inhibits T-cell activation
Etanercept (Busca et al. 2007)	Fusion protein that inactivates TNF- α
Extracorporeal photopheresis (Abu-Dalle et al. 2014)	Involves exposing leukocytes to a photosensitizing drug and ultraviolet A light that leads to immunomodulatory activity
Fecal microbiota transplantation	Re-establishes gut microbiota (NCT04711967, NCT04280471, NCT03819803, NCT05094765, NCT03812705 and NCT04769895)
Infliximab (Couriel et al. 2004)	Monoclonal antibody that blocks TNF- α activity
Mesenchymal stromal cells (Kurtzberg et al. 2020)	Immunomodulatory activity (NCT05333029, NCT04744116 and NCT04629833)
Mycophenolate mofetil (Pidala et al. 2010; Furlong et al. 2009)	Inhibits IMPDH proliferation of T and B cells
Pentostatin (Ragon et al. 2018; Bolanos-Meade et al. 2005)	Inhibits adenosine deaminase and induces lymphocyte apoptosis
Sirolimus (Hoda et al. 2010; Pidala et al. 2009)	Inhibitor of mTOR and inhibits T-cell proliferation and differentiation
Tocilizumab (Roddy et al. 2016; Drobyski et al. 2011; Ganetsky et al. 2019)	Anti-IL6 receptor antibody and blocks IL-6, a proinflammatory cytokine
Vedolizumab (Li et al. 2022) 22	Inhibits activation of $\alpha 4\beta 7$ integrin and trafficking of T cells to the GI mucosa

Abbreviations: *ATG* antithymocyte globulin, *CNI* calcineurin inhibitor, *IMPDH* inosine-5'-monophosphate dehydrogenase

had high NRM rates due to subsequently developing lower GI SR-aGVHD (Hartwell et al. 2017; Levine et al. 2015; Major-Monfried et al. 2018). Consequently, the MAGIC aGVHD algorithm has been developed into a commercially available test for predicting severe aGVHD, treatment resistance, and NRM (Eurofins-Viracor n.d.).

While Minnesota (clinical) and MAGIC (biomarker) risk categorizations both identify patient groups more or less likely to respond to CS, and with higher or lower NRM rates, outside of clinical trials, these approaches have not been established as standard of care. Perhaps the biggest concern relates to inevitably exposing a proportion of patients who are deemed high-risk to intensified IST who did not stand to benefit due to imperfect performance characteristics of these risk categories.

6.5 Gastrointestinal (GI) GVHD

Approximately 50% of patients with aGVHD have GI tract involvement making it the second most affected organ behind skin (MacMillan et al. 2015); upper (UGI) and mid-to-lower GI tract (LGI) may be involved separately or together. Approximately two-thirds of the GI involvement (or 30% overall) is UGI, typically manifesting as persistent nausea, vomiting, and anorexia beyond the time when conditioning toxicities should have resolved (Nikiforow et al. 2018). UGI aGVHD without LGI involvement typically responds well to systemic CS $\pm$ oral topical CS, such as beclomethasone dipropionate and budesonide, and has a lower risk for NRM (Nikiforow et al. 2018; Hockenbery et al. 2007). LGI aGVHD is characterized by watery, secre-

tory, protein-rich diarrhea >500 mL per day. With advanced aGVHD, the entire GI tract becomes inflamed and edematous, leading to voluminous diarrhea, malabsorption, mucosal ulceration, sloughing, bleeding, and severe abdominal pain (Naymagon et al. 2017). NRM for Stage 3–4 LGI aGVHD is higher than for stage 1–2 (MacMillan et al. 2015). While <10% present with de novo stage 3–4 LGI aGVHD, the vast majority who develop severe disease do so while already on first-line therapy. Although rates of severe GI GVHD have declined over the past decades, severe GI GVHD remains a major cause of post-HCT mortality (Harris et al. 2016; Aasebo et al. 2022). Standard first-line therapy remains a PED of 2 mg/kg/day, with second-line IST only being added when manifestations are deemed SR-aGVHD. Outcomes are poor for these patients who do not completely respond to high-dose CS; accordingly, reducing aGVHD mortality is contingent upon improving responses in patients with lower GI GVHD.

Vedolizumab recently emerged as a potential preventative option for intestinal GVHD. It is a humanized monoclonal antibody that inhibits $\alpha 4\beta 7$ interaction on T cells with their ligand MAdCAM-1 on endothelium within the GI tract and associated lymphoid tissues thereby inhibiting trafficking of donor T lymphocytes to these target tissues (Floisand et al. 2019). This mechanism of action supported the evaluation of vedolizumab added to standard GVHD prophylaxis after unrelated HCT in an international double-blind, randomized, placebo-controlled phase 3 trial (Chen et al. 2024). The primary endpoint of lower-GI aGVHD-free survival by day+180 after HCT was met: 85.5% (79.2–90.1) with vedolizumab versus 70.9% (63.2–77.2) with placebo (hazard ratio, 0.45; 95% confidence interval, 0.27–0.73; $P < 0.001$) (Chen et al. 2024).

The role of vedolizumab for GI aGVHD treatment is yet to be defined. A meta-analysis that included seven studies showed vedolizumab to yield a pooled ORR of 61% at 14 days after its initiation (pooled risk ratio, RR = 14.14, 95% CI: 2.95–67.71), 50% at 28 days (pooled RR = 7.36, 95% CI: 2.14–25.37), and 77% at 12 months (pooled RR = 13.66, 95% CI: 3.50–53.35) (Li et al. 2022), yet OS remained low with pooled OS

of 34% at 6 months and 29% at 12 months. Significantly increased risks such as infections, impaired metabolism, and nutritional complications occurred in the treatment but not prophylactic setting. Further research is required to delineate its role in GI aGVHD treatment.

6.6 Future Directions in Gastrointestinal (GI) GVHD

Strategies beyond pharmacological immunosuppressants are under investigation that recognize the relative importance of the intestinal microbiome and mucosal barrier in immune-mediated inflammatory diseases like GI GVHD. We learned from murine models that butyrate, a short chain fatty acid, was decreased in intestinal epithelial cells (IECs) of mice experiencing GVHD (Goldberg et al. 2003). Restoring butyrate levels by increasing intestinal butyrate-producing bacteria reduced experimental aGVHD severity and mortality (Goldberg et al. 2003). Administration of resistant potato starch (RPS) to normal healthy human volunteers promotes increased butyrogenic bacteria and intestinal butyrate levels (Couriel et al. 2005). A subsequent feasibility study where patients (age ≥ 10 years) were given RPS from D–7 to D+100 after HCT showed feasibility of RPS administration and RPS significantly increasing salutary intestinal butyrate levels (Gomez-Almaguer et al. 2008). A phase 2 placebo-controlled study is underway to see if aGVHD rates can be reduced (NCT02763033). Interestingly, this approach might not be wise if treating non-intact GI mucosa after GI aGVHD onset, based on other observations where increased butyrogenic bacteria were associated with subsequent SR-GVHD or cGVHD (Hamilton et al. 2024), reminding us that context can be important.

Another prophylactic approach involves manipulation of the gut microbiome by capsular administration of a commercially produced, uniform bacterial consortia investigational product (IP), before and after HCT (Couriel et al. 2004). The IP was designed to decolonize GI pathogens, improve epithelial barrier integrity, and induce immune tolerance to prevent blood stream infec-

tions, antimicrobial resistance, and other pathogen-associated negative clinical outcomes. Data from a small, yet randomized Phase Ib study of adults demonstrated engraftment of bacterial strains in the IP; blood stream infection rates were 10% for the IP vs 43% with placebo (OR: 0.15; 95% CI: 0.01, 1.13, $p = 0.04$), representing a 77% relative risk reduction; systemic antibiotic exposure, and incidence of fever and neutropenia were also lower with the IP (NCT04995653) (Harper and McMillen 2021).

FMT for treatment and prophylaxis of aGVHD has also been evaluated in patients but the best source of FMT product remains unclear (Harris et al. 2016). Screened third party FMT is commonly used and can be delivered as oral capsules. In one placebo-controlled phase II study of HCT recipients or AML patients undergoing induction chemotherapy, FMT timed with neutrophil recovery was safe and ameliorated intestinal dysbiosis but did not meet its primary end-point of reducing all-cause infection rate (Hartwell et al. 2017). However, a *post-hoc* analysis of the FMT treated group showed a lower rate of grade II-IV aGVHD in patients who had greater donor microbiota engraftment, suggesting that FMT may protect against aGVHD (Hashmi et al. 2015). Subsequently, a placebo-controlled phase 2 double blind trial of oral encapsulated third-party FMT to prevent severe aGVHD is ongoing at Fred Hutchinson Cancer Center (FHCC) (NCT06026371) (Hashmi et al. 2018). Lingering concerns with third-party FMT are the transfer of drug-resistant microorganisms that can be life-threatening (Hills et al. 2019). Large, randomized studies of FMT or FMT-like (i.e., bacterial consortia) will be required before these therapies can be offered routinely.

When intestinal crypt loss is present, sometimes with a denuded mucosal barrier, GI recovery can be exceedingly slow despite potentially effective novel IST. Lithium carbonate can attenuate GVHD via effects on the intestinal stem cell niche (Cutler et al. 2021). It is thought to do this by inhibiting the normal glycogen synthase-3 (GSK-3) antagonism of WNT/b-catenin signaling which then drives cell proliferation and epithelial regeneration; fortunately, long-term

lithium use is not associated with increased risk of colorectal cancer (Hockenbery et al. 2007). In a pilot study at FHCC, adults with GI GVHD and mucosal denudation at endoscopy received oral lithium plus second-line IST; 50% had CR by intention to treat, but 8 of 12 who started lithium (with therapeutic drug monitoring) ≤ 3 days from baseline endoscopy showing denuded mucosa had a CR and survived >1 year. Compared to historical controls, 2-year survival was 35% vs 10% (Hoda et al. 2010). While fatigue, dulled sensorium, and flattened affect occurred commonly this was difficult to assess in a nonrandomized critically ill cohort.

Lastly, there has been recent interest in teduglutide and apraglutide, which are both GLP-2 analogs. GLP-2 helps maintain and protect GI epithelial cells, including enterocytes, intestinal stem cells, and Paneth Cells, which are direct targets of aGVHD (Deeg 2007; Holler et al. 2014). Teduglutide is approved for once daily SQ injection in patients age ≥ 1 year with short bowel syndrome (SBS). A pediatric case series with pre- and post-treatment correlative immunohistochemistry data suggested efficacy in 3 children with refractory GI aGVHD (Deeg et al. 1995). Apraglutide is a once weekly SQ injectable that has undergone successful phase 3 testing in adults with SBS with US regulatory submission expected in 2025. Results of a proof-of-concept phase 2 trial (NCT05415410) where apraglutide was added to ruxolitinib as second-line treatment for SR-lower-GI aGVHD in patients age ≥ 12 years showed apraglutide was well-tolerated with encouraging efficacy in a severely ill population (Cuvelier et al. 2022).

6.7 Nutritional Support in Acute GI GVHD

In addition to IST, aGVHD of the GI tract management includes medical nutrition therapy and supportive care for nausea, early satiety, vomiting, dysphagia, anorexia, diarrhea, abdominal pain, and bleeding to effectively prevent and/or manage malnutrition (Table 6.3). Malnutrition results from inadequate intake compounded by

Table 6.3 Medical nutrition therapies and supportive care in GI GVHD (Harper and McMillen 2021; Akbulut and Yesildemir 2021)

Nutrition impact symptom	Medical nutrition therapy
Xerostomia	Diet plan that includes highly moist foods Normal saline rinses Saliva stimulants (e.g., citric acid drinks, sugar-free candies, mints, gum, cevimeline, or pilocarpine) Dry mouth products (e.g., Biotene (rinse/spray/gel), Xerostom (rinse/spray/gel), mouth Kote (rinse), adhesive tablets (Zylimelts), or adhesive strips (Xerostrrips) Gastronomy tube for EN if symptoms preclude adequate calorie-protein intake for weight maintenance and support
Dysgeusia	Diet plan that includes a variety of spicy or flavorful foods (as tolerated) Promotion of good oral hygiene
Nausea/vomiting	Diet plan that includes salty or sour foods, cold and clear liquids, small frequent meal portions, low-fat and low-fiber foods Antiemetics Starting PN or postpyloric EN if symptoms are severe
Anorexia	Consume smaller food portions at more frequent intervals Consumption of high-calorie snacks
Dysphagia	Referral to a speech pathologist for swallowing evaluation Esophageal dilatation for severe symptoms Gastronomy tube for EN if symptoms preclude adequate calorie-protein intake for weight maintenance and support
Diarrhea	Antidiarrheal medications Low lactose diet Avoid GI irritants/simulants and irritating disaccharides Diet plan that includes low-fat, low-insoluble fiber Discontinuing oral intake for large volume, watery diarrhea >500 mL/day
Pancreatic enzyme insufficiency or atrophy	Screening tests (fecal elastase or serum trypsinogen) to confirm fat malabsorption Oral pancreatic enzyme replacement therapy Dietary fat restriction Supplementation with products containing medium-chain triglycerides

Abbreviations: *EN* enteral nutrition, *GI* gastrointestinal, *PN* parenteral nutrition

increased metabolic demands, alterations in protein-calorie intake, nutrient absorption, and utilization leading to macronutrient (e.g., protein catabolism or protein-losing enteropathy) and micronutrient (e.g., electrolytes, vitamin D, B12, zinc) deficiencies, all which can be exacerbated by systemic inflammation and medication toxicities (van der Meij et al. 2013; McMillen et al. 2021; Harper and McMillen 2021). Because malnutrition is a well-established factor associated with poor survival in HCT (Deeg et al. 1995; Beckerson et al. 2019), registered dietitians with HCT expertise are critical to any aGVHD management team. Formal initial and longitudinal nutrition assessments are necessary to determine patients' needs for diet modifications and/or nutrition support (enteral (EN) and/or parenteral nutrition (PN). Although patients with intractable nausea, severe abdominal pain, diarrhea, and/or

overt GI bleeding may be placed on bowel rest when they meet the criteria for having a nonfunctional GI tract, oral intake (or EN) should be resumed as soon as possible to prevent gut atrophy (August et al. 2009). Adequate EN during the early transplantation course has been associated with reduced early NRM, improved survival, and GVHD-free/relapse-free survival at 5 years compared to PN and inadequate nutrition (Beckerson et al. 2019). EN protects against overall and gut aGVHD (Zama et al. 2021) perhaps because of its ability to maintain mucosal integrity, modulate the immune response to the effects of intensive chemo/radiotherapy, and support intestinal microbiota homeostasis after HCT (Beckerson et al. 2019; Arai et al. 2015; Seguy et al. 2012).

A healthy (diverse) gut microbiota plays a key role in the immune system, with links to both reduced aGVHD and GVHD-related mortality

(Burgos da Silva et al. 2022; Jenq et al. 2015; Holler et al. 2014). This is not surprising because GVHD-associated disruption of GI mucosal integrity decreases nutrient assimilation and promotes systemic infection through bacterial translocation and enteropathy (Seguy et al. 2012). As maintaining a variety of enteral nutrients is essential to the gut microbiota diversity (Hills Jr. et al. 2019), having a variety of food choices plus good safety practices is more important than a “neutropenic” or “low microbial” diet, which is not supported by evidence in this patient population (Arends et al. 2017). It follows that while PN might be the only option for patients with the most severe manifestations of GI aGVHD or inability to maintain placement of an enteral feeding tube (Arends et al. 2017), if PN is initiated, early reintroduction of oral intake should iteratively be assessed because it leads to improved nutritional outcomes (Imataki et al. 2006).

Whereas the importance of oral intake (or EN) in HCT is well established, the role of micronutrients, their requirements in HCT, and their association with HCT outcomes is less clear (Segon et al. 2022). Vitamins A and D both appear to have immunomodulatory roles, including suppression of host antigen-presenting cells and donor T-cell activation, inhibition of proinflammatory cytokines, and increases in regulatory T cells (Chen and Mayne 2018). These preclinical findings are supported by emerging clinical data—children with lower vitamin A levels had more frequent and more severe intestinal aGVHD (Lounder et al. 2017) which led to a randomized, double-blinded, placebo-controlled phase II trial that compared 4000 IU/kg vitamin A versus placebo given within one week before conditioning (Khandelwal et al. 2024). The cumulative incidence of GI aGVHD and cGVHD was lower in the vitamin A-treated group in the “as-treated” analysis.

The role of vitamin D has been more variable with studies either showing a positive or no association of vitamin D deficiency with GVHD incidence and/or severity [reviewed in Rodriguez-Gil et al. (2022)]. Until these results are validated in well-designed randomized studies, clinicians should use clinical judgment to provide supple-

mentation where appropriate. If micronutrient deficiencies are found or suspected before or after transplantation, a multivitamin (without iron) supplement may be considered. Additionally, EN and PN formulations should be selected carefully to ensure appropriate vitamin and mineral provision.

6.8 Chronic GVHD

Chronic GVHD remains the leading cause of late post-HCT morbidity and mortality (Lee et al. 2003; Pidala et al. 2011). It is a polymorphic, multiorgan syndrome with clinical manifestations having inflammatory and/or fibrotic characteristics but often characterized by the latter as the syndrome gains chronicity. Experimental models suggest that the syndrome arises after early onset inflammation and host tissue damage from pretransplant conditioning and stimulation of alloreactive donor T cells (Zeiser and Blazar 2017b). Subsequently, thymic injury, loss of peripheral tolerance, emergence of alloreactive B and T (B/T) cells, and loss of regulatory B/T cells together lead to chronic inflammation. The final common pathways appear to involve activated IL-17 and colony-stimulating factor-1 (CSF-1)-dependent, donor-derived macrophages that produce profibrotic cytokines, ultimately leading to the deposition of extracellular collagen by activated fibroblasts (Alexander et al. 2014; MacDonald et al. 2017a, b). The incidence of cGVHD ranges from 30% to 70% (Arai et al. 2015; Arora et al. 2016; Bachier et al. 2021), with increasing rates observed in the early 2000s, attributed to increasing recipient age and the use of unrelated donor and peripheral blood stem cell grafts (Arai et al. 2015). However, recent data suggests that cGVHD is decreasing among adults, children, and patients who underwent HCT for either malignant or nonmalignant disease (Carpenter et al. 2024).

Before the initial National Institutes of Health (NIH) consensus criteria of 2005 (Filipovich et al. 2005), a cGVHD diagnosis was challenging because of its clinical heterogeneity, the coexistence of aGVHD manifestations, and ambiguities in traditional classification

approaches (Lee et al. 2003; Shulman et al. 1980). Assignment of cGVHD severity was similarly challenging because of the lack of validated tools to measure cGVHD in individual organs. The 2005 NIH cGVHD consensus effort harmonized methods and definitions for diagnosis, organ scoring, and global severity assignment, as well as treatment response criteria for clinical trials. Importantly, late aGVHD (after day 100) could occur with classic cGVHD manifestations, in which case it was termed overlap subtype, or could occur alone, in which case it has been associated with worse overall and failure-free survival (Holtan et al. 2016). Furthermore, *mild*, *moderate*, and *severe* global severity scores were associated with progressively worse OS and increased NRM (Arai et al. 2011). Iterative refinements in the 2014 NIH criteria (Jagasia et al. 2015) provided clarifications and more reliable tools that were used in a clinical trial that led to the first FDA approval of a cGVHD therapy (Miklos et al. 2017; Jaglowski and Blazar 2018). The third NIH consensus (Pavletic et al. 2021) reported cGVHD basic research and clinical practice gaps relevant to early detection and had the goal to avoid highly morbid cGVHD forms by detecting them (hopefully) before advanced fibrosis causes irreversible morbidity. While these NIH efforts have been undoubtedly pivotal, together with progress in understanding cGVHD biology, several key challenges remain in modern cGVHD management.

6.9 Treatment Challenges in Chronic GVHD

Aside from prevention, cGVHD therapy might ideally rebalance a dysregulated immune system and reverse and prevent fibrosis, leading to the complete resolution of all cGVHD manifestations. This would also be accomplished without significant patient toxicities and would ultimately enable durable and complete withdrawal of IST. While this may be possible for a minority of patients, treatment success might also be considered if there is a reduction in symptom burden, preserved activities of daily living, and improved

quality of life. The ideal agent remains elusive, but the cGVHD treatment paradigm has clearly shifted from baseline long-term broadly active CS-based IST toward agents that target specific pathways involved in cGVHD pathogenesis.

6.9.1 First-Line Therapy

Systemic treatment with IST has been used to control moderate-to-severe cGVHD, and so far, multiple phase 3 studies (Sullivan et al. 1988; Koc et al. 2000, 2002; Arora et al. 2001; Martin et al. 2009; Gilman et al. 2012; Carpenter et al. 2018; Miklos et al. 2023) have not identified more efficacious first-line therapy than CS beginning at the PED of 0.5–1 mg/kg/day $\pm$ calcineurin inhibitor (CNI) although inclusion of CNI led to less avascular necrosis in a prospective phase 3 study (Koc et al. 2002). After $\sim$ 14 days, a taper usually can begin and per results of one prospective study, the average reaches 0.2–0.25 mg/kg/day (or 0.4–0.5 mg/kg every other day) within 3 months after starting treatment (Martin et al. 2009). Roughly half of transplant centers prescribe alternate-day prednisone tapers based on old literature, primarily case series in non-HCT populations, supporting fewer side effects and lower total cumulative prednisone exposure. A randomized phase 2 study of daily versus alternate-day prednisone for cGVHD in children was deemed a high-priority trial at the third State of the Science Symposium but did not move forward and so this question remains unanswered for all age groups (Appelbaum et al. 2015). In clinical practice, when cGVHD is not well controlled on an alternating day regimen, some patients can be better controlled simply by switching to an equivalent-dose daily regimen, possibly reflecting the shorter biological half-life of prednisone in some patients. In another scenario, blood sugar levels may swing more erratically in diabetic patients who are on alternate-day versus daily prednisone regimens.

Eventually, after cycles of attempted prednisone tapering and dose reescalation, the dose reaches 2.5 mg/day (5.7 mg/m²/day of hydrocortisone) in adults and 1.2–2.2 mg/m²/day ($\sim$ 5–9 mg/m²/day of hydrocortisone) in children

and adolescents, which approximates daily cortisol production (Esteban et al. 1991; Linder et al. 1990). At this point, in non-Cushingoid patients, prednisone is often converted to a split-dose regimen of hydrocortisone (two-thirds taken at ~8 AM and one-third taken at 3–5 PM; no later than 6 PM) to facilitate adrenal recovery during the final taper. Formal testing for secondary adrenal insufficiency is often done when a patient remains Cushingoid on low-dose prednisone and if confirmed, the final taper generally is much slower. Moreover, GVHD activity may recur even when tapering below *physiological* PED, which may also dictate length of the final prednisone (or hydrocortisone) taper.

6.9.2 Prednisone Dependence, Refractoriness, Unresponsiveness, or Intolerance

Phase 3 trials have shown that by 6 months into therapy, only ~50% of patients had a CR or partial response (PR) and just over 25% had a CR. As a result, insufficient response to first-line therapy or recurrent cGVHD after tapering initial therapy means that >50% of patients receive ≥ 3 lines of therapy and remain on IST for a median of 2–3.5 years (Bachier et al. 2021; Lee et al. 2018); for a minority of patients, time to discontinue all IST may extend beyond 7 years. Indications for second-line therapy include worsening cGVHD manifestations in a previously affected organ, development of signs and symptoms of cGVHD in a previously unaffected organ or absence of improvement after 1 month of standard primary treatment (*refractory*), inability to decrease prednisone below 1 mg/kg per day within 2 months (*dependent*), or significant treatment-related toxicity (*intolerance*). Providers must be aware that some organs are irreversibly damaged (*unresponsive*) like lacrimal or salivary glands or fascia (contractures).

Beyond second-line therapy, additional patience is required before concluding treatment failure. For example, manifestations like sclerosis might be responding even if there is no

improvement by 3 months if the prior sclerosis disease trajectory has plateaued, and prednisone can be tapered meaningfully. Well-documented, methodical, and serial skin assessments are necessary to be fully certain of this determination. The recently FDA-approved, more targeted cGVHD therapeutics discussed below had a median time to first response ranging from 4 to 12 weeks, $\geq 50\%$ ORR at or through 24 weeks, and generally even higher best ORRs. Therefore, in the absence of clear disease progression or intolerance, a reasonable trial is 6 months before declaring efficacy failure.

Finally, it can be helpful for patients to set expectations regarding the duration of cGVHD therapy. Whereas solid organ transplant recipients must take IST for life, a registry study suggested that 32% of HCT recipients *durably* stop IST by 10 years after cGVHD diagnosis, noting that just under one-half were ineligible to achieve this goal due to competing risks of relapse or NRM (Chen et al. 2023). Several factors can shorten the time to durably discontinue IST, with recipients of cord blood, naive T-cell depletion, or post-transplant cyclophosphamide developing cGVHD and/or morbidity less frequently and discontinuing IST much sooner.

6.9.3 Steroid-Refractory cGVHD

Until 2017, there was no standard of care for second-line therapy but commonly used were ECP (Abu-Dalle et al. 2014), low-dose methotrexate (Giaccone et al. 2005; Huang et al. 2005), mycophenolate mofetil (Lopez et al. 2005), rituximab (Kharfan-Dabaja et al. 2009), sirolimus (Couriel et al. 2005; Johnston et al. 2005; Jurado et al. 2007), imatinib (Arai et al. 2016), pento-statin (Goldberg et al. 2003; Jacobsohn et al. 2007), and low-dose interleukin-2 (IL-2) (Koreth et al. 2016) based on mostly heterogeneous, uncontrolled retrospective or prospective study data. The aforementioned NIH consensus effort, which included important clinical trial endpoints, has now facilitated FDA approval of four targeted SR-cGVHD therapeutics (ibrutinib, ruxolitinib, belumosudil, and axatilimab).

Ibrutinib selectively and irreversibly inhibits Bruton's tyrosine kinase (BTK) and interleukin-2-inducible T-cell kinase (ITK), which are important for signaling and activation of B cells and T cells (Miklos et al. 2017). Its 2017 FDA approval was based on the phase Ib/II study results of 42 subjects with SD/SR-cGVHD. Notably, subjects had to have at least an erythematous rash or cGVHD with oral erythema. The best ORR was 67% (21% CR) with a median time to response of 12.4 weeks (range, 4.1 to 42.1 weeks), and ~ 71% of responses were sustained for ≥ 20 weeks. High response rates were seen in skin (88%), mouth (88%), and GI tract (91%). Eleven of 18 patients with baseline sclerosis showed sclerotic improvement. Patient counseling about side effects should include mention of fatigue, diarrhea, muscle spasms, and bruising. The latter is thought to be due to impaired platelet aggregation and while no major hemorrhages were reported in GVHD trials, the product label advises holding ibrutinib for ± 7 days around major surgery and ± 3 days around minor surgery. Arrhythmias including atrial fibrillation deserve mention but occurred infrequently in GVHD trials. In 2022, the ibrutinib FDA cGVHD indication was revised to age ≥ 1 year, based on the results of the prospective multicenter iMAGINE study. The study determined pediatric dosing, found no new safety signals, and showed a similar 6-month ORR and a numerically higher best ORR compared to the pivotal adult trial (Carpenter et al. 2022).

Ruxolitinib selectively inhibits Janus-associated kinase (JAK) 1 and 2, thereby inhibiting the JAK-signal transducer and activator of the transcription (JAK/STAT) pathway. Because the activation of intracellular JAK 1/2 leads to transcription of proinflammatory cytokines, mediation of inflammatory neutrophil migration, and upregulation of major histocompatibility class II expression on antigen-presenting cells (Zeiser and Blazar 2017b), JAK/STAT inhibition has opposite effects, as well as diminishing alloreactive T-cell infiltration into target organs mediated by IFN- γ and CXCR3. In 2021, the FDA-approved

ruxolitinib based on the results from REACH3, a phase 3, randomized controlled trial (Zeiser et al. 2021), which compared the efficacy and safety of ruxolitinib versus the best available therapy (BAT) in patients with SD or SR-cGVHD. ORR at 24 weeks was 50% for ruxolitinib versus 26% for BAT ($p \leq 0.001$). ORR was 53% for the LGI tract, 50% for the esophagus, 50% for the mouth, 41% for the skin, and 40% for the UGI tract (40%). Responses were ~3-fold higher for ruxolitinib, and partial responses (PR) were more likely in both arms. Patients who crossed over from BAT to ruxolitinib after week 24 subsequently still had a 50% ORR (95% CI: 37.8, 62.2) by 24 weeks, with the best ORR reaching 81.4% (95% CI: 70.3, 89.7), and 74.3% of these being PRs (Zeiser et al. 2023). Ruxolitinib led to longer median failure-free survival than BAT (hazard ratio, 0.37; $P < 0.001$), higher symptom response (odds ratio, 2.62; $P = 0.001$), and spared CS slightly better than BAT. Reversible anemia and thrombocytopenia were more common in the ruxolitinib versus BAT arm; these can be managed with dose reductions and supportive care. Bacterial and viral infections occurred with similar frequency in both arms, but fungal infections were numerically higher in the ruxolitinib arm (11.5% vs. 5.7%). Real-world retrospective studies evaluating ruxolitinib in SR-cGVHD found similar best ORRs and adverse events (AE) (Redondo et al. 2022; Ferreira et al. 2021; Wu et al. 2021; Zeiser et al. 2015), and median times to response were 3 weeks to 2 months (range, 0.5–8 months), which is helpful when counseling patients (Redondo et al. 2022; Zeiser et al. 2015).

Belumosudil (KD025) selectively inhibits rho-associated coiled-coil-containing protein kinase-2 (ROCK2). Because ROCK2 activation results in sequential upregulation of STAT3 phosphorylation and then elaboration of Th17-specific transcription factors, ROCK2 inhibition leads to downregulation of proinflammatory cytokines including IL-17 and IL-21 in T cells (Zanin-Zhorov et al. 2014), with a shift toward more regulatory T cells. ROCK2 activation also results in the polymerization of globular actin to fibrous actin (Riches et al. 2015; Knipe et al. 2015), which in turn leads to further expression

of profibrotic genes, promoting myofibroblast and collagen production, in a profibrotic environment. Preclinical cGVHD models showed that inhibition of ROCK2 by belumosudil interferes with this profibrotic process. In 2021, the FDA-approved belumosudil based on the results of ROCKstar, the phase II, open-label, randomized, multicenter study that compared belumosudil 200 mg once daily versus 200 mg twice daily in patients who had received 2–5 previous lines of cGVHD therapy (Cutler et al. 2021). Because the best ORR, at a median follow-up of 14 months, was similar (74% vs. 77%) without differences in individual organ response rates, the recommended dose is 200 mg once daily for age ≥ 12 years, unless on a proton pump inhibitor, in which case 200 mg twice daily is recommended. Despite being a more heavily pretreated study population, ORRs were similar if previously treated with ruxolitinib (68%) or ibrutinib (74%); similar to ibrutinib and ruxolitinib, most responses were PR. The median time to response was 5 weeks (range, 4–66 weeks), and the median duration of response was 54 weeks. Belumosudil appeared to provide clinically meaningful benefit in patients with ≥ 4 involved organs, in patients who were refractory to their last line of treatment, and in patients with severe cGVHD. CS dose decreased by a median of 45% from baseline in most patients; 31% discontinued CS therapy, and 22% discontinued CNI therapy over 12 months. Belumosudil was generally well tolerated with a 12% AE-attributable discontinuation rate. The most common AEs of any grade were fatigue, diarrhea, nausea, cough, and upper respiratory tract infection. Reactivation of cytomegalovirus infection was not observed.

Axatilimab inhibits colony-stimulating factor-1 receptor (CSF-1R) on monocytes and macrophages and blocks receptor activation by CSF1 and interleukin-34 (IL-34) (Kitko et al. 2023; Ordentlich et al. 2016), thereby playing key roles in regulating tissue macrophages. CSF1R signaling-dependent monocytes and macrophages are essential mediators of inflammation and fibrosis in cGVHD and a range of

autoimmune diseases (Alexander et al. 2014; Jardine et al. 2020). Preclinical studies showed that early post-transplantation antibody blockade of CSF1R could target donor-derived macrophages and ameliorate cGVHD (Alexander et al. 2014). After a successful initial phase I/II clinical SR-cGVHD study, axatilimab showed promising preliminary efficacy and safety results; preferential elimination of CSF1R-dependent nonclassical monocytes and tissue macrophages was demonstrated (Kitko et al. 2023). Efficacy and safety of axatilimab were further established in AGAVE 201, the pivotal phase 2, multinational study that preceded its FDA approval in 2021 (Wolff et al. 2024). A total of 241 patients were randomized (1:1:1) to 0.3-mg, 1-mg, and 3-mg dose groups. While all dose groups met the primary end point, the 6-month ORR was best (74%) for the 0.3-mg group (95% CI: 63–83), 67% (95% CI, 55–77) for the 1-mg group, and 50% (95% CI, 39–61) for the 3-mg group. At least 5-point improvements on the modified Lee Symptom Scale were reported in 60%, 69%, and 41% of the patients in the three dose groups, respectively. In parallel, AE-attributable discontinuations were only 6% for the 0.3-mg group versus 22% for the 1-mg group and 18% for the 3-mg group. Patients should be counseled about dose-dependent transient laboratory abnormalities related to CSF1R blockade (elevated AST, ALT, CPK, LDH, amylase, and lipase) and periorbital edema, albeit this was infrequent at the approved intravenous dose of 0.3 mg every 2 weeks.

6.10 Tailoring Therapy in Chronic GVHD

Although the FDA approvals of ibrutinib, ruxolitinib, belumosudil, and axatilimab reflect an improved understanding of cGVHD biology, harmonized diagnostic, response criteria, and clinical trial endpoints, the existence of these multiple treatment options prompts a questioning of how best to sequence next-in-line therapy for SR-cGVHD. A corollary question is whether

there is a preferred agent for specific organ involvement (e.g., oral vs. skin vs. lung) or clinical phenotype (e.g., inflammatory or fibrotic manifestations). In practice, FDA-labeling indications largely direct the answer to question number one, but the corollary question was never addressed by any of these pivotal clinical trials. While ibrutinib and ruxolitinib are FDA-approved for the failure of ≥ 1 systemic therapy, the use of belumosudil or axatilimab requires the failure of ≥ 2 lines of therapy. In Europe, the European Medicines Agency (EMA) has similarly approved ruxolitinib and ibrutinib; however, belumosudil and axatilimab have orphan drug designations. The recognition that ruxolitinib is the only drug approved based on having a comparator treatment arm of the best available therapy may influ-

ence some providers to select ruxolitinib before other treatments.

Beyond FDA labeling and pivotal trial study design, selection of next-in-line therapy often considers the agent's mechanism of action, underlying disease indication for HCT (e.g., ruxolitinib with a history of myelofibrosis or ibrutinib with CLL), and relevance of its AE profile to an individual patient's comorbidities inclusive of infections, availability of an oral formulation (e.g., axatilimab being only intravenous), feasibility of patient co-pays, provider experience, and patient preference (Table 6.4). Taking all the above, next-in-line therapy commonly involves shared clinical decision-making.

Establishing patient-tailored approaches to cGVHD treatment would require validated bio-

Table 6.4 Selection of treatment considerations

	Ibrutinib (Imbruvica (ibrutinib) [package insert] 2024)	Ruxolitinib (Jakafi (ruxolitinib) [package insert] 2021)	Belumosudil (Rezurock (belumosudil) [package insert] 2024)	Axatilimab (Niktimvo (axatilimab) [package insert] 2024)
FDA indication	Refractory cGVHD after ≥ 1 LOT in patients age 1 year and older	Refractory cGVHD after ≥ 1 LOT in patients age 12 years and older	Refractory cGVHD after ≥ 2 LOTs in patients age 12 years and older	Refractory cGVHD after ≥ 2 LOTs in patients with weight ≥ 40 kg
Mode of administration	Oral	Oral	Oral	IV
Most common ($\geq 20\%$) non-laboratory AEs of all grades	Fatigue, bruising, diarrhea, stomatitis, nausea, muscle spasms, hemorrhage, and pneumonia	Infections	Infections, asthenia, edema, nausea, diarrhea, abdominal pain, dyspnea, cough, hemorrhage, hypertension. MSK pain, headache	Infections, MSK pain, fatigue, nausea, headache
Most common ($\geq 10\%$) grade ≥ 3 laboratory AEs	Neutropenia	Anemia, neutropenia, thrombocytopenia, hypercholesterolemia, elevated ALT, GGT and lipase	Elevated GGT, lymphocytes decreased	None
Common AEs leading to discontinuation of agent	Fatigue, pneumonia	Pneumonia	Nausea	Unknown
Other considerations	Avoid with other strong CYP3A inhibitors (e.g., CSA)	Avoid with fluconazole doses >200 mg/day	Twice daily dosing requiring when co-administered with PPI (may increase patient OOP costs)	Monitor for infusion-related reactions, including fever, chills, Rash, flushing, dyspnea, and hypertension

Abbreviations: AEs adverse events, CSA cyclosporin, GGT gamma glutamyl transferase, IV intravenous, LOT line of therapy, MSK musculoskeletal, OOP out-of-pocket, PPI proton pump inhibitor

markers. In the future, predictive biomarkers might assist with the sequencing of CS-sparing therapies, but this information is currently lacking. Although not formally validated, in practice, for some patients, serum creatine kinase (and/or aldolase) can serve as helpful response biomarkers for patients with the atypical cGVHD manifestation of myositis (Couriel et al. 2002; Shahzad et al. 2022). Beyond anecdotal case series, validated biomarkers are not in routine clinical use (Bidgoli et al. 2022; Ji et al. 2023). To address this need, the Chronic GVHD Consortium initiated a prospective, observational cohort study in June 2020, “Predicting the Quality of Response to Specific Treatments” (Hamilton et al. 2024) (NCT04431479), to identify clinical and biological predictors of treatment response among patients receiving first-, second-, or third-line systemic therapy for cGVHD. Detailed clinical data pertaining to cGVHD phenotype and activity and whole blood samples are collected at defined timepoints with the hope of defining clinical and biological subgroups of patients and their response to treatments.

6.11 Should Chronic GVHD Therapies Be Added or Substituted?

The argument for *substituting* rather than simply *adding* next-in-line therapy is that to date, no controlled first-line therapy trials have demonstrated superior efficacy of two-versus-one CS-sparing agent(s) with some trials showing worse outcomes (Sullivan et al. 1988; Koc et al. 2000, 2002; Arora et al. 2001; Martin et al. 2009; Gilman et al. 2012; Carpenter et al. 2018; Miklos et al. 2023). Adding IST is also concerning simply because infection with or without cGVHD is a major cause of NRM, with the hazard for late fatal infections in one large registry study being ~4-fold for adults with cGVHD while still on IST and ~9-fold for children (Norkin et al. 2019; DeFilipp et al. 2021). These data have informed the predominant FHCC practice to substitute next-in-line therapy usually by discontinuing the failed CS-therapeutic within 2 weeks from the substitution.

Whether substituting should still be the preferred approach in the era of recent FDA-approved therapies is arguably best addressed in clinical trials. Combining these newer therapeutics has gained attention because they might have greater CS-sparing potential and be better suited for combination, ideally in first-line therapy with little or no CS. Each of the four approved agents has demonstrated overall responses in SR-cGVHD but most of the responses were partial. Only ibrutinib has been evaluated in first-line therapy, but like the above-mentioned controlled studies, the ORR to ibrutinib plus prednisone was no better than prednisone plus placebo (Miklos et al. 2023). Until new single-agent CS-sparing studies are reported, combining new-era CS-therapeutics should be done sparingly because supporting data is limited to case series mostly in the context of cGVHD that was refractory to multiple therapies (Caputo et al. 2024; Chin et al. 2024; Pusic et al. 2024). In summary, while cGVHD pathophysiology continues to be elucidated, it is logical to study new-era CS-agent combinations that target different biological pathways, with minimal overlapping toxicities, in preclinical models and well-designed prospective trials.

6.12 Early Detection and Preemption

The third NIH cGVHD consensus (Pavletic et al. 2021) highlighted the goal to avoid highly morbid, advanced fibrotic forms of cGVHD through early clinical detection supplemented with relevant validated biomarkers. Feasibility of earlier cGVHD recognition requires multiple stakeholders, including nontransplant providers (e.g., primary care provider, referring hematologist/oncologist), patients, and caregivers. At FHCC, patients and their local health providers have phone or email access to a team of survivorship-dedicated support staff, including long-term follow-up (LTFU) telemedicine registered nurses who answer questions and triage questions needing additional expertise to an LTFU Attending physician (Lee et al. 2021). Coordinating care with the patient’s original transplant center is ideal; however, a directory of GVHD specialists

(BMTinfonet.org [n.d.](#)) accepting patients regardless of where the patient received their transplant is available.

Encouraging patients and their caregivers to be proactive in cGVHD monitoring might also facilitate early diagnosis and intervention and validated tool development to accomplish this is ongoing. Meanwhile, patient and caregiver education about key reportable signs and symptoms of cGVHD is recommended before discharge from the transplant center back to the referring provider.

Beyond early symptom reporting by patients, clinic visits by a provider with cGVHD expertise should incorporate methodical, iterative (every 1–3 months) organ-specific screenings during the first 1–2 years post-HCT to facilitate early cGVHD diagnosis, perhaps even in the absence of symptoms (Kitko et al. [2021](#)). Pre-HCT baseline examination of the skin, joint range of motion, eyes, oral cavity, lungs, and genitals is important to document baseline tissue abnormalities that may confound subsequent cGVHD interpretations. At these visits, cGVHD symptom screening can be efficiently delegated to a clinical staff member; any positive sign or symptom can be made readily available to the expert provider. Electronic health (e-Health) tools like the “eGVHD app,” which has been used worldwide, can facilitate cGVHD assessment and diagnosis because it encourages systematic patient physical evaluations and decreases diagnosis and scoring errors (Schoemans et al. [2016, 2018a, c](#)).

Susceptibility or (subclinical) diagnostic biomarkers might further aid in the early diagnosis

of cGVHD but are still in the research phase (Bidgoli et al. [2022](#); Kitko et al. [2021](#); Martens et al. [2023](#)). Finally, whether subsequent preemptive therapy for the earliest clinical manifestations of cGVHD can prevent highly morbid forms of cGVHD is now being examined in clinical trials (NCT05996627).

6.13 Highly Morbid Forms and Atypical Manifestations

Highly morbid cGVHD manifests when irreversible fibrosis leads to sequelae that negatively impact physical functioning, quality of life, and survival; particularly common organ syndromes include severe dry eyes, bronchiolitis obliterans, and sclerotic skin and joint/fascia involvement (Wolff et al. [2021](#)). Less commonly, very severe oral and genital manifestations occur, and more rarely, a wasting syndrome not explained by GI causes occurs. Atypical cGVHD manifestations outside the eight commonly affected cGVHD organs (eyes, mouth, skin, liver, lung, GI and GI tracts, joints/fascia) include the hematopoietic system (immune-mediated cytopenias), endothelium, serous membranes, central and peripheral nervous systems, musculoskeletal system, and kidneys (Table [6.5](#)) [reviewed in Cuvelier et al. ([2022](#))].

Prevention and treatment of highly morbid or atypical manifestations remain problematic because of incomplete understanding of their biology, under-recognition, relative infrequency of atypical manifestations, lack of predictive and

Table 6.5 Target organs and manifestations in atypical versus NIH-defined cGVHD

Atypical target organs and manifestations	
Central nervous system	Cognitive deficits, meningoencephalitis, demyelinating diseases, CNS vasculitis
Peripheral nervous system	Neuropathy, myasthenia gravis
Lungs	COP, nonspecific interstitial pneumonia, PPFE
Serositis	Pericardial effusion, pleural effusion, ascites
Renal	Proteinuria, nephrotic syndrome, tubular, glomerular, or interstitial disease, vascular disease
Hematologic	Autoimmune hemolytic anemia, immune thrombocytopenia purpura, autoimmune neutropenia
Muscles, fascia, joints	Edema, muscle cramps, arthralgia, arthritis, myositis

Adapted from Cuvelier et al. ([2022](#))
Abbreviations: COP cryptogenic organizing pneumonia, PPFE pleuroparenchymal pulmonary fibroelastosis

diagnostic biomarkers, and lack of well-established, efficacious organ-specific treatment approaches. Clinical trials focused on organ-specific treatments are additionally challenging to conduct because of the protean nature of cGVHD and the small number of patients with single-organ involvement of interest.

Upon recognition of cGVHD, collaborative involvement of relevant subspecialists can be beneficial. Colleagues in pulmonology, ophthalmology, gynecology, dermatology, nephrology, and rheumatology are often consulted for their subspecialty expertise and may provide additional management insights based on their experience with either GVHD or analogous immune-mediated conditions. Methodical care coordination is required when multiple subspecialists are involved, and it is important that the transplant team remains central to ensure that patients are receiving optimal cGVHD management, including supportive and ancillary care, as well as surveillance for graft function, relapse (if malignant HCT indication), and late effects.

6.14 Ancillary and Supportive Care in Chronic GVHD

The cGVHD burden cannot be overemphasized given that manifestations may evolve and/or recur over several years, yet never in isolation from other HCT late effects. Late effects result from varying degrees of interaction between pre-transplant exposures, post-transplant complications, and positive or negative influences of age, gender, genetics, and lifestyle. Given that clinical immunological tolerance may take years rather than months, toxicities arising from IST and medications used to treat side effects of IST can contribute to chronic health conditions per se, for example, hypertension, dyslipidemia, and chronic renal insufficiency to name a few. It follows that comprehensive LTFU care involves not only monitoring and management of cGVHD, but also monitoring for late effects [reviewed in Carpenter (2023)]. However, a major challenge is that LTFU attenuates over time and compliance with LTFU guidelines is suboptimal by survivors

and healthcare providers especially when patients are adolescents or young adults, transplanted for standard-risk malignancy or nonmalignant disease, and perceived to be doing well (Miyamura et al. 2019; Armenian et al. 2012; Khera et al. 2011). An American Society of Transplantation and Cellular Therapy (ASTCT) survey found that respondents agreed on allo-HCT survivors having needs separate from GVHD and that complications can arise during transitions of care from pediatric to adult or from transplant center to primary medical provider. However, only 45% had dedicated LTFU clinics; 84% of individual practitioners preferred to provide their own survivorship care. Other challenges include a lack of subspecialty LTFU expertise, logistics of the LTFU model of care, and financial issues (Hashmi et al. 2015, 2018). More work is needed to optimize LTFU, particularly for young adult survivors who are more prone to becoming lost to follow-up.

Two key components of supportive care for cGVHD include optimizing nonsystemic IST (usually topically directed) for individual organ sites, and broad infection prophylaxis and timely revaccination [reviewed in Carpenter and Englund (2016)] given that a registry study found infection to be a primary or contributing factor for ~30% of late deaths (Norkin et al. 2019) and another large study (Foord et al. 2020) showed vaccine-preventable late infections were >30-fold more frequent than in the general population. Baseline antimicrobial prophylaxis directed against shingles, pneumocystis *jirovecii* pneumonia, encapsulated organisms, and often molds is recommended during cGVHD therapy. Routine post-transplant vaccinations are recommended and detailed elsewhere (Carpenter and Englund 2016; Rubin et al. 2014; Reynolds et al. 2023). Immunoglobulin replacement therapy is prescribed judiciously per ASTCT “Choosing Wisely” unless profoundly low IgG (often with immeasurable IgA) (Bhella et al. 2018).

Additional challenges can be medication burden and monitoring for complications of polypharmacy and noncompliance, plus frequent care coordination around issues of mood disturbance, and ensuring that there is adequate psychosocial

support for patients and caregivers; referral to appropriate subspecialists who can assist with symptom control or potentially help with multifactorial problems that often can include contribution(s) from other HCT late effects.

Survivors are at risk for depression, fatigue, pain, social withdrawal, educational problems, and vocational problems including unemployment and dependent living (Armenian et al. 2012). Among allogeneic recipients, cGVHD severity, being female, and having current pain were associated with depression and fatigue (Jim et al. 2016). A registry study showed that return to work by adolescent young adults improved over time but many became or remained unemployed or on medical disability (Bhatt et al. 2021). Return-to-work programs are unavailable at most centers despite 87% endorsing this as a problem (Salit et al. 2020), and ASTCT-endorsed return-to-work guidance for providers of HCT is available in Open Access and can be used as a tool to counsel and support these patients (Salit et al. 2022).

6.15 Future Directions and Conclusions

Remarkable progress has been made in our understanding of GVHD biology, and we are better able to categorize, diagnose and grade GVHD manifestations, and capture cGVHD symptoms. Clinical frameworks now exist to facilitate the design and implementation of clinical trials, as well as increased industry and government collaborations. Novel therapies are more available after failure of initial therapy and additional investigational agents are undergoing evaluation. Challenges still remain, with the priority being the study of first-line agents that hopefully will spare chronic CS exposure. More effective treatments for severe GI aGVHD and highly morbid forms of cGVHD will still be needed when GVHD prevention or early therapy approaches fail. Whether patient-tailored over one-size-fits-all approaches can be achieved remains a stretch goal. Serum biomarkers are more developed for clinical use in aGVHD but

still not widely utilized in practice, and cGVHD biomarkers to assign risk, predict, diagnose, and assess treatment response remain in development.

Artificial intelligence-based, machine-learning algorithms may have a role in the management of GVHD, given the preliminary data showing its potential to predict aGVHD and cGVHD (Martens et al. 2023; Chandra et al. 2024; McNeil et al. 2022; Arai et al. 2019), as well as to recognize cutaneous cGVHD (McNeil et al. 2022).

Despite these challenges, the future of GVHD appears bright. Collaborations between transplant clinicians, community providers, and subspecialists are growing stronger to provide optimal GVHD care, as academia, industry, government, and patient advocacy groups also work together toward eradicating this debilitating HCT complication.

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